The running example for the metaplectic Jacquet–Langlands series: $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4 - \zeta_3)_K$

David Joyner Claude (Anthropic)

November 8, 2026 — draft

Abstract

We fix one concrete case of the standing assumptions of Parts I–IV and [JC], and evaluate in it the main objects of each part. These are the cocycle parameters, the commutator pairings, the good elements, the constants b_D and c_D, the wild toral decompositions, and the global normalization ledger. Every statement is either checked here, or quoted with its location and status label. The aim is to make the general constructions concrete, and to give each part a common place to test its formulas.

Theorem numbers. References to the series are to the drafts in the repository: Parts I and II, Draft 8; Part III, Draft 9; Part IV, Draft 5; and the rank-two toral positivity note.

1 The data

Base field	$K = \mathbb{Q}(\zeta_3)$, totally imaginary, $\mu_3 \subset K$
Covering degree, rank	$n = 3, \quad r = 2, \quad (n, r) = 1$
Parameters (Part I, Assumption 1.1)	$r' = 2, \quad c = 1, \quad c_v = r'c = 2$
Checks	$3 \mid r - 1 + 2c = 3, \quad 3 \mid r - 1 + 2rc_v = 9$
Division algebra	$D = (-1, 4 - \zeta_3)_K, \quad 4 - \zeta_3 = (1 - \zeta_3)(3 + \zeta_3)$
Ramification	$S = \{\mathfrak{p}_3, \mathfrak{p}_7\}, \quad \mathfrak{p}_3 = (1 - \zeta_3), \quad \mathfrak{p}_7 = (3 + \zeta_3)$
Cyclic subfield (Part I, [JC])	$L = K(i) = \mathbb{Q}(\zeta_{12}) \subset D$

Fact 1 ([**P**] Ramification)**.** D is ramified exactly at $\mathfrak{p}_3$ and $\mathfrak{p}_7$.

Proof. At $\mathfrak{p}_3$ and $\mathfrak{p}_7$. The element -1 is a nonsquare in the residue fields $\mathbb{F}_3$ and $\mathbb{F}_7$, since $3 \equiv 7 \equiv 3 \pmod 4$. The element $4 - \zeta_3$ has valuation 1 at each of these places. So $(-1, 4 - \zeta_3) = -1$ at both.

At the other finite odd places. Both entries are units, so the symbol is 1. This includes the other prime above 7, namely $(3 + \zeta_3^2)$, since 7 splits in K.

At 2. The product formula forces the symbol to be 1.

Machine check. Sage gives:

```
K.<z> = CyclotomicField(3); a, b = K(-1), 4 - z
for p in [2,3,5,7,13]:
    for P, e in K.ideal(p).factor():
        print(p, P, K.hilbert_symbol(a, b, P))
```

This returns -1 exactly at $(z - 1)$ and at $(-3z - 2)$, and $+1$ at (2), (5), $(3z + 1)$, $(z + 4)$ and $(z - 3)$. Here $(-3z - 2) = (3 + z)$, since $(-3z - 2)/(3 + z) = z^2$ is a unit. □

The hypotheses of Part I also hold.

- $L = K(i)$ is cyclic of degree $r = 2$ and lies in D.
- At both places of S, L_v is a field, because -1 is not a square in $\mathbb{F}_3$ or $\mathbb{F}_7$.
- $|S| = 2$ is even, as it must be for $r = 2$.

2 The local fields

Place	K_v	Type for $n = 3$	$\lvert 3\rvert_v$	$\lvert K_v^\times / K_v^{\times 3}\rvert$
$\mathfrak{p}_3 \in S$	$\mathbb{Q}_3(\zeta_3)$ ($e = 2$, $q = 3$)	wild	$1/9$	81
$\mathfrak{p}_7 \in S$	$\mathbb{Q}_7$ (contains μ_3)	tame	1	9
(2)	$\mathbb{Q}_2(\zeta_3)$ ($f = 2$, $q = 4$)	tame, split for D	1	9
$(3 + \zeta_3^2)$, $(13) = \mathfrak{p}\mathfrak{p}'$, …	$\mathbb{Q}_\ell$, $\ell \equiv 1$ (3)	tame, split	1	9
complex	$\mathbb{C}$	cover splits	9	1

The last column is $n^2|n|_v^{-1}$ at the finite places (Part IV, Proposition 4.2). Locally at $\mathfrak{p}_3$ we have $3 + \zeta_3 \equiv 1 \pmod{\mathfrak{p}_3}$, which is a square. So $D_{\mathfrak{p}_3} \cong (-1, \varpi)$ with $\varpi = \zeta_3 - 1$; this is exactly the model $E_0 \oplus E_0 j$, $j^2 = \varpi$, of Part IV, Example 6.3. At $\mathfrak{p}_7$, $D_{\mathfrak{p}_7} \cong (-1, 7)_{\mathbb{Q}_7}$.

3 Part I: cocycle, commutators, good elements

- *Diagonal torus* (Part IV, Proposition 2.1). With $c_v = 2$,

$$[\tilde{t}, \tilde{u}] = \prod_{i,j} (t_i, u_j)_3^{A_{ij}}, \qquad A = -I + 5J \equiv \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \pmod 3,$$

  so $[\tilde{t}, \tilde{u}] = (t_1/t_2, u_1/u_2)_3$. A regular $\mathrm{diag}(t_1, t_2)$ is good if and only if t_1/t_2 is a cube.
- *Quadratic tori* (Part I, Proposition 3.1 and Example 3.7). Here $[x, y] = (Nx, Ny)_F^2 (x, y)_E^{-1}$. The good elements are $F^\times E^{\times 3}$ (Part I, Proposition 3.5; Part IV, Theorem 3.1).
- *Index of the good elements in a quadratic torus*, $|3|_v^{-1}$ (Part I, Theorem 3.9):
  - at $\mathfrak{p}_3$ it is 9: the covered torus is a Heisenberg group, and its genuine irreducibles have dimension 3;
  - at $\mathfrak{p}_7$ it is 1: every element is good, and the covered torus is abelian.
- *Sign.* The matching sign $(-1)^{r-1}$ of Part I, Theorem 4.5, is -1.
- *A regular-but-not-good element* (Part IV, Remark 3.3). The element $\mathrm{diag}(1, \zeta_3)$ is regular. It is not good at $\mathfrak{p}_3$, because $\zeta_9 \notin \mathbb{Q}_3(\zeta_3)$. Its cube is the identity.

4 Part III: the transform L_D

	$b_D = \lvert 3\rvert^{-1/2}$	$c_D = \lvert 3\rvert^{-3/2}$	$R_E = \Delta(t)/\Delta(t^3)$
$\mathfrak{p}_7$ (tame)	1	1	$\equiv 1$ (Part III, Lemma 7.4)
$\mathfrak{p}_3$ (wild)	3	27	$\in \{1, 3, 9\}$ (below)

- *At $\mathfrak{p}_7$.* L_D is a dimension-preserving bijection of irreducible characters (Part III, Theorem 7.6) **[E]**.
- *At $\mathfrak{p}_3$, divisibility.* The cover does not split, and every genuine irreducible has dimension divisible by 3 (Part III, Proposition 8.1) **[E]**.
- *At $\mathfrak{p}_3$, the prediction.* If L_D maps irreducibles to irreducibles, then $\dim \tau = 27 \dim \rho$. In particular $L_D\mathbf{1}$ would be an irreducible of dimension 27.

5 Part IV: the wild place $\mathfrak{p}_3$

The three quadratic tori

$F = \mathbb{Q}_3(\zeta_3)$ has three quadratic extensions. All of them embed in $D_{\mathfrak{p}_3}$. On $E^1 \cong E^\times/F^\times$, Part IV, §6.3 and the toral note give the following **[P]**.

E/F	values of R_E	R_E as a permutation character	coefficients
unramified $F(i)$	$9,\ 1$	$1 + 2\,\mathrm{Reg}(C_4)$	$(3,2,2,2)$
ramified $F(\sqrt{u\varpi})$, either type	$9,\ 3,\ 1$	$1 + \mathrm{Reg}(C_2) + \mathrm{Reg}(C_6)$	$(3,2,1,1,1,1)$

Both expansions have value $9 = |3|_F^{-1}$ at the identity. So the restriction of $L_D\mathbf{1}$ to every covered quadratic torus is an actual genuine character of dimension $3 \cdot 9 = 27$, consistent with the prediction above. In the cyclic algebra model $j^2 = \varpi$:

t	j	$1+j$	$1+\varpi j$
$R(t)$	1	3	9
$(L_D\mathbf{1})(t^*)$	3	9	27

What is open in the example

[C] *Is $L_D\mathbf{1}$ at $\mathfrak{p}_3$ the character of an irreducible genuine representation of $\widetilde{D^\times_{\mathfrak{p}_3}}$, of dimension 27?*

- By the positivity lemma (Part IV referee report on Draft 5, §2), it suffices that every $L_D\Theta_\rho$ be positive definite.
- Torus positivity is now known.
- The missing input is the restriction to the p-elementary subgroups of the covered pro-3 group.

This is the smallest instance of the wild integrality problem.

6 Global: good classes and the normalization ledger

- *Good classes* (Part IV, Theorem 5.2) **[P]**. A regular $\gamma \in D^\times(K)$ with $E = K(\gamma)$ has a rational lift that is good everywhere if and only if $\gamma \in K^\times E^{\times 3}$. The map $[s] \mapsto [s^3]$ is a bijection onto these classes.

- *The ledger* (Part IV, Example 8.8). Take $\lambda_v = 1$ at the finite split places. Then

$$\Lambda = \lambda_\infty \cdot b_{D,\mathfrak{p}_3} \cdot b_{D,\mathfrak{p}_7} = 3\lambda_\infty,$$

  and $\Lambda = 1$ forces $\lambda_\infty = 1/3 = |3|_\infty^{-1/2}$.

- *The identity-term heuristic.* It gives the complex factor $\lambda_\infty |3|_\infty^{-1} = 1/27$, which balances $c_{D,\mathfrak{p}_3} = 27$ by the product formula.

- *Unramified places.* FK's unit matching needs $p \nmid n$ and $(n, N_2) = (3, 1) = 1$. Both hold at every finite place outside S, since $\mathfrak{p}_3 \in S$ is the only place above 3.

7 [JC]

In [JC] the lifting runs from genuine automorphic representations of an n-fold cover of $L^\times$, here with $L = \mathbb{Q}(\zeta_{12})$, to ε-invariant genuine representations of the cover of $D^\times$. Part I, §1, says this. A section of [JC] evaluated in this example would complete the picture.

Suggested use

Each part could contain a short subsection, "The running example", pointing to the relevant section of this note. Each new formula in a draft can then be tested here first. The open question of §5 is the natural first target for the wild integrality problem.